EXPLICIT EVALUATION OF SPRUNG'S HALF-LOGARITHM H-MATRIX AT $X = 0$ FOR $p = 2$ SUPERSINGULAR REDUCTION

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Abstract. We give an explicit evaluation of Sprung's half-logarithm matrix $\mathcal{H}(X)$ at $X = 0$ for elliptic curves with supersingular reduction at $p = 2$. The matrix limit defining $\mathcal{H}(0)$ degenerates to the closed form $A^{-2}B$, where A is the Frobenius action matrix and B encodes the eigenvalues of the characteristic polynomial of Frobenius. As a consequence, the constant terms of the Sprung half-logarithms in their canonical basis are rational over $\mathbb{Q}$, although they a priori take values in the local quadratic extension $\mathbb{Q}_2(\alpha)$. We further give an explicit bridge ε^{-1} connecting Sage's Dieudonné basis to Sprung's canonical basis, and verify it bit-exactly on 180 elliptic curves drawn from the Cremona database with conductors in the range [37, 63 096]. The rationality phenomenon is interpreted as a renormalisation of the otherwise divergent Pollack half-logarithms at $T = 0$, in the spirit of [3, Remark 6.16].

1. Introduction

Iwasawa theory for elliptic curves at supersingular primes was reformulated by Kobayashi [1], building on the construction of Pollack's plus/minus p-adic L-functions [2], for the case $a_p = 0$. The case $a_p \neq 0$ was treated by Sprung [3], who introduced a pair of half-logarithms $\log_\alpha^\sharp$, $\log_\alpha^\flat$ and a pair of p-adic L-functions $L_p^\sharp$, $L_p^\flat$ satisfying the canonical decomposition

$$L_p(E, \alpha, X) = L_p^\sharp(E, X) \cdot \log_\alpha^\sharp(X) + L_p^\flat(E, X) \cdot \log_\alpha^\flat(X). \tag{1}$$

The matrix $\mathcal{H}(X) = (\log_\alpha^\sharp, \log_\alpha^\flat)^T$ is defined as the projective limit

$$\mathcal{H}(X) := \lim_{n\to\infty} C_1(X) \cdot C_2(X) \cdots C_n(X) \cdot A^{-(n+2)} \cdot B, \tag{2}$$

where $C_k(X) = \begin{pmatrix} a_p & 1 \\ -\Phi_{p^k}(1+X) & 0 \end{pmatrix}$, A is the Frobenius action matrix at $X = 0$, and B encodes the eigenvalues α, β of $X^2 - a_pX + p$. The convergence of this limit and its analytic properties have been studied in [3, 4], but explicit closed-form evaluations are scarce.

The case $p = 2$ is delicate: Sprung's main theorem in [4] is stated for *odd* supersingular primes, leaving the $p = 2$ case partly open in the literature. At the same time, computational software such as Sage's `padic_lseries.Dp_valued_series` [7] works with a slightly different basis convention, related to Sprung's by

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an ε-bridge $\varepsilon = (I - \varphi)^2$ where φ is the Frobenius matrix in the Dieudonné basis $(\omega, \varphi\omega)$.

This note presents two observations.

(I) Triviality of the limit at $X = 0$. The defining limit (2) degenerates algebraically at $X = 0$: since $\Phi_{p^k}(1) = p$ for all $k \geq 1$, the cofactor matrices $C_k(0)$ coincide with A, and $\mathcal{H}(0) = A^{-2}B$ holds independently of the truncation level n. This is Theorem 3.1 below.

(II) The Sage–Sprung bridge. The conversion between Sage's Dieudonné basis and Sprung's canonical basis is realised by an explicit rational matrix ε^{-1}. We give the explicit form in Theorem 3.3 and verify it bit-exactly on 180 elliptic curves.

These results are elementary, but useful as a working tool for computational Iwasawa theory at $p = 2$. They allow researchers to convert Sage's output to Sprung's convention without ambiguity, and they exhibit the constant components of Sprung's half-logarithms at $X = 0$ as elements of $\mathbb{Q}^2$, in spite of their a priori realisation in $\mathbb{Q}_2(\alpha)$.

Notation and conventions. Throughout, $p = 2$ and $E/\mathbb{Q}$ is an elliptic curve without complex multiplication with good supersingular reduction at p, equivalently $a_p(E) \in \{-2, 0\}$. We write α, β for the eigenvalues of Frobenius acting on the p-adic Tate module, with characteristic polynomial $X^2 - a_p X + p$. We denote by $K_\alpha := \mathbb{Q}_2(\alpha)$ the local quadratic extension. The Frobenius matrix in the Dieudonné basis $(\omega, \varphi\omega)$ is taken to be

$$\varphi = \begin{pmatrix} 0 & -1/p \\ 1 & a_p/p \end{pmatrix}, \tag{3}$$

following the convention of Sage's implementation. The ε-bridge is

$$\varepsilon := (I - \varphi)^2 \in M_2(\mathbb{Q}). \tag{4}$$

Acknowledgments. The numerical verifications in Section 4 were carried out in SageMath 10 with the full Cremona database. The author thanks the SageMath developer community and J. E. Cremona for maintaining the elliptic-curve infrastructure on which this work depends.

2. Setting and preliminaries

2.1. **The Sprung half-logarithm matrix.** Sprung's matrix $\mathcal{H}(X)$ is defined by (2). We recall the auxiliary matrices: at $p = 2$,

$$A = \begin{pmatrix} a_p & 1 \\ -p & 0 \end{pmatrix} = \begin{pmatrix} a_p & 1 \\ -2 & 0 \end{pmatrix}, \tag{5}$$

$$B = \begin{pmatrix} -1 & -1 \\ \beta & \alpha \end{pmatrix}. \tag{6}$$

The cyclotomic polynomials $\Phi_{p^k}(t)$ satisfy $\Phi_2(1) = 2$, $\Phi_4(1) = 2$, and more generally $\Phi_{2^k}(1) = 2$ for all $k \geq 1$ (the only prime dividing the order of a primitive root). Thus at $X = 0$:

$$\Phi_{p^k}(1) = p, \qquad k \geq 1. \tag{7}$$

2.2. **The ε-bridge.** In computational settings, p-adic L-functions and their decompositions (1) are usually computed in Sage's Dieudonné basis $(\omega, \varphi\omega)$. The bridge to Sprung's canonical basis is given by $\varepsilon = (I - \varphi)^2$.

Lemma 2.1. $\det(\varepsilon) = \dfrac{|E(\mathbb{F}_p)|^2}{p^2} = \dfrac{(p+1-a_p)^2}{p^2}$.

Proof. A direct computation gives

$$\det(I - \varphi) = (1) \cdot \left(1 - \frac{a_p}{p}\right) - \left(\frac{1}{p}\right) \cdot (-1) = \frac{p+1-a_p}{p}.$$

Squaring and applying $|E(\mathbb{F}_p)| = p + 1 - a_p$ yields the stated formula. $\square$

In our setting $p = 2$, equation (8) becomes:

$$\det(\varepsilon) = \frac{(3-a_p)^2}{4} = \begin{cases} 25/4 & \text{if } a_p = -2, \\ 9/4 & \text{if } a_p = 0. \end{cases} \tag{8}$$

3. Main results

Theorem 3.1 (Triviality of the limit at $X = 0$)**.** *Let p be any prime and let A and B be as in Section 2.1. The Sprung half-logarithm matrix evaluated at $X = 0$ is*

$$\mathcal{H}(0) = A^{-2} \cdot B, \tag{9}$$

algebraically closed without nontrivial limit; in fact, for every $n \geq 1$ the truncated product satisfies

$$C_1(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^{-2}. \tag{10}$$

Proof. By (7), $C_k(0) = A$ for every $k \geq 1$. Hence $C_1(0) \cdots C_n(0) = A^n$, and

$$C_1(0) \cdots C_n(0) \cdot A^{-(n+2)} = A^n \cdot A^{-(n+2)} = A^{-2},$$

independently of n. Passing to the limit and multiplying by B yields $\mathcal{H}(0) = A^{-2}B$. $\square$

The matrix $A^{-2} \cdot B$ has explicit entries in $K_\alpha = \mathbb{Q}_2(\alpha)$. For $p = 2$:

Proposition 3.2 (Explicit $\mathcal{H}(0)$)**.** *At $p = 2$ we have:*

(i) *For $a_p = -2$ ($\alpha = -1 + i$, $\beta = -1 - i$):*

$$\mathcal{H}(0) = \begin{pmatrix} -i/2 & i/2 \\ \frac{1}{2} - \frac{i}{2} & \frac{1}{2} + \frac{i}{2} \end{pmatrix}, \qquad \det \mathcal{H}(0) = -\tfrac{i}{2}. \tag{11}$$

(ii) *For $a_p = 0$ ($\alpha = i\sqrt{2}$, $\beta = -i\sqrt{2}$):*

$$\mathcal{H}(0) = \begin{pmatrix} 1/2 & 1/2 \\ \frac{i\sqrt{2}}{2} & -\frac{i\sqrt{2}}{2} \end{pmatrix}, \qquad \det \mathcal{H}(0) = -\tfrac{i\sqrt{2}}{2}. \tag{12}$$

Proof. For $a_p = -2$ one has $\det A = 2$ and $A^{-2} = \begin{pmatrix} -1/2 & 1/2 \\ -1 & 1/2 \end{pmatrix}$. The product $A^{-2}B$ with $B = \begin{pmatrix} -1 & -1 \\ -1-i & -1+i \end{pmatrix}$ yields the stated matrix. The case $a_p = 0$ is analogous. $\square$

Theorem 3.3 (Sage–Sprung bridge). *Let $E/\mathbb{Q}$ have $p = 2$ supersingular reduction. The conversion between Sage's basis $(\omega, \varphi\omega)$ and Sprung's canonical basis is the linear map*

$$u_{\text{Sprung}} = \varepsilon^{-1} \cdot u_{\text{Sage}}, \qquad \varepsilon = (I - \varphi)^2. \tag{13}$$

For the constant components of the Sprung half-logarithms in Sage's basis (with the normalisation of [3, §5]), this yields the rational pairs:

(i) $a_p = -2$:

$$u_{\text{Sage}} = (7/4,\ 3/4), \qquad u_{\text{Sprung}} = (4/5,\ 9/10) \in \mathbb{Q}^2. \tag{14}$$

(ii) $a_p = 0$:

$$v_{\text{Sage}} = (1,\ 2), \qquad v_{\text{Sprung}} = (-2/3,\ 4/3) \in \mathbb{Q}^2. \tag{15}$$

Proof. At $a_p = -2$ one computes

$$\varepsilon = \begin{pmatrix} 1/2 & 3/2 \\ -3 & 7/2 \end{pmatrix}, \qquad \varepsilon^{-1} = \begin{pmatrix} 14/25 & -6/25 \\ 12/25 & 2/25 \end{pmatrix}, \qquad \det \varepsilon = \tfrac{25}{4}.$$

Direct multiplication gives $\varepsilon^{-1} \cdot (7/4,\ 3/4)^T = (4/5,\ 9/10)^T$. At $a_p = 0$ one similarly has

$$\varepsilon = \begin{pmatrix} 1/2 & 1 \\ -2 & 1/2 \end{pmatrix}, \qquad \varepsilon^{-1} = \begin{pmatrix} 2/9 & -4/9 \\ 8/9 & 2/9 \end{pmatrix}, \qquad \det \varepsilon = \tfrac{9}{4},$$

and $\varepsilon^{-1} \cdot (1,\ 2)^T = (-2/3,\ 4/3)^T$. □

Corollary 3.4 (Rationality). *The constant components of Sprung's half-logarithms at $X = 0$ in the canonical basis are elements of $\mathbb{Q}^2$, although Sprung's matrix $\mathcal{H}(0)$ takes a priori values in $K_\alpha = \mathbb{Q}_2(\alpha)$. Specifically:*

$$u_{\text{Sprung}}|_{a_p=-2} = (4/5,\ 9/10), \qquad v_{\text{Sprung}}|_{a_p=0} = (-2/3,\ 4/3). \tag{16}$$

4. Numerical verification

We verified Theorem 3.3 on 180 elliptic curves drawn from the Cremona database [8] using SageMath [7]. The sample is partitioned as follows: 75 curves with $a_p(E) = -2$ in the conductor range $[37,\ 997]$, and 105 curves with $a_p(E) = 0$ in the conductor range $[77,\ 63\,096]$. The $a_p = 0$ block is subdivided into 75 low-conductor curves in $[77,\ 225]$ and 30 curves in $[1\,001,\ 63\,096]$ obtained via logarithmically spaced bucket sampling. All curves have $\operatorname{rank}(E/\mathbb{Q}) \geq 1$.

For each curve we computed the Sage Dieudonné Frobenius matrix φ, formed $\varepsilon = (I - \varphi)^2$, and applied ε^{-1} to $(7/4,\ 3/4)^T$ if $a_p = -2$ or $(1,\ 2)^T$ if $a_p = 0$. In every case the result equalled the predicted Sprung-basis vector $(4/5,\ 9/10)^T$ or $(-2/3,\ 4/3)^T$ respectively, bit-exactly in rational arithmetic over $\mathbb{Q}$.

4.1. Sample of verified curves.

Table 1 shows a representative sample of 12 curves selected to span the full conductor range.

Label	Conductor	$a_p(E)$	rank	Bridge match
`37a1`	37	-2	1	✓
`389a1`	389	-2	2	✓
`571b1`	571	-2	2	✓
`5077a1`	5 077	-2	3	✓
`17963c1`	17 963	-2	2	✓
`53461b1`	53 461	-2	3	✓
`77a1`	77	0	1	✓
`1001a1`	1 001	0	1	✓
`15855a1`	15 855	0	2	✓
`30487a1`	30 487	0	3	✓
`39812b1`	39 812	0	2	✓
`63096b2`	63 096	0	1	✓

TABLE 1. Sample of curves verifying Theorem 3.3. Each row reports $\varepsilon^{-1} \cdot u_{\mathrm{Sage}} = u_{\mathrm{Sprung}}$ bit-exactly in $\mathbb{Q}^2$.

4.2. **Computational details.** The verification was conducted in SageMath 10 using the full Cremona database [8] (largest conductor 499 998). Stored rank information was used in lieu of `E.rank()` for performance. The total compute time was approximately five minutes on a standard workstation. The verification code and the complete log files are publicly available at

https://github.com/relkuch-del/sprung-h-matrix-p2.

Class	n	Conductor range	Bit-exact
$a_p = -2$ supersingular	75	[37, 997]	75/75
$a_p = 0$ supersingular (low)	75	[77, 225]	75/75
$a_p = 0$ supersingular (high)	30	[1 001, 63 096]	30/30
Total	**180**	[37, 63 096]	**180/180**

TABLE 2. Summary of numerical verification. Zero mismatches and zero errors over four orders of magnitude in the conductor range.

4.3. **Summary.**

5. DISCUSSION: CONSISTENCY WITH POLLACK'S HALF-LOGARITHMS

In [3, Remark 6.16], Sprung relates his half-logarithms to Pollack's plus/minus logarithms $\log_2^{\pm}$ in the case $a_p = 0$:

$$\log_\alpha^\sharp = -\tfrac{1}{2}\log_2^+ \cdot \alpha, \qquad \log_\alpha^\flat = \log_2^- . \tag{17}$$

At $X = 0$, however, Pollack's logarithms behave singularly. Recall [2, Lemma 4.1]:

$$\log_2^+(T) = \frac{1}{2}\prod_{n\geq 1}\frac{\Phi_{2n}(1+T)}{2}, \qquad \log_2^-(T) = \frac{1}{2}\prod_{n\geq 1}\frac{\Phi_{2n-1}(1+T)}{2}. \tag{18}$$

Evaluating at $T = 0$:

- $\log_2^-(0) = 0$ *structurally*, since $\Phi_1(1) = 0$ enters the product;

- $\log_2^+(0)$ does *not* converge in $\mathbb{Q}_2$: each truncation N has 2-adic valuation $\leq -N$, with $v_2(\log_2^+(0)|_N) \to -\infty$ as $N \to \infty$.

Thus the constant values of Pollack's half-logarithms at $T = 0$ are not well-defined in $\mathbb{Q}_2$. The role of Sprung's normalisation factor $A^{-(n+2)}$ in (2) is precisely to absorb this divergent behaviour into a well-defined element of $\mathbb{Q}^2$. Theorem 3.1 makes this concrete at $X = 0$: the renormalisation collapses to multiplication by A^{-2}, and the resulting matrix $A^{-2}B$ has rational entries in Sprung's canonical basis.

This structural consistency *explains* the rationality observed in Theorem 3.3 and Corollary 3.4: the vectors $u_{\text{Sprung}}, v_{\text{Sprung}} \in \mathbb{Q}^2$ are precisely the renormalised limits of the otherwise divergent Pollack constants.

6. Open questions

We close with three natural questions raised by, but not answered in, the present note.

Convergence of $\mathcal{H}(X)$ for $X \neq 0$. The analysis here treats only the special point $X = 0$, where the limit degenerates algebraically. For $X \neq 0$ the cofactor matrices $C_k(X)$ depend nontrivially on k, and the convergence of the projective limit (2) requires more delicate input. One expects an explicit description of $\mathcal{H}(X)$ as a power series in X over a suitable Banach algebra, extending the techniques of [3, 4]; this is left to future work.

The Iwasawa main conjecture at $p = 2$. The main theorem of [4] concerns odd supersingular primes. The case $p = 2$ is still open in the published literature. The bridge ε^{-1} provided by Theorem 3.3 clarifies the relation between explicit Sage computations and Sprung's theoretical framework, and may be one of the technical ingredients for an eventual treatment of the $p = 2$ case.

Higher-order components. The vectors $u_{\text{Sage}}, v_{\text{Sage}}$ in Theorem 3.3 are the constant components of Sage's p-adic L-series at $p = 2$. The full series expansions of $\log_\alpha^\sharp(X)$ and $\log_\alpha^\flat(X)$ around $X = 0$ display further rationality phenomena which warrant a separate investigation.

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